



Efficiently Simulate Quantum Gates Validating with Quantum-inspired KAN

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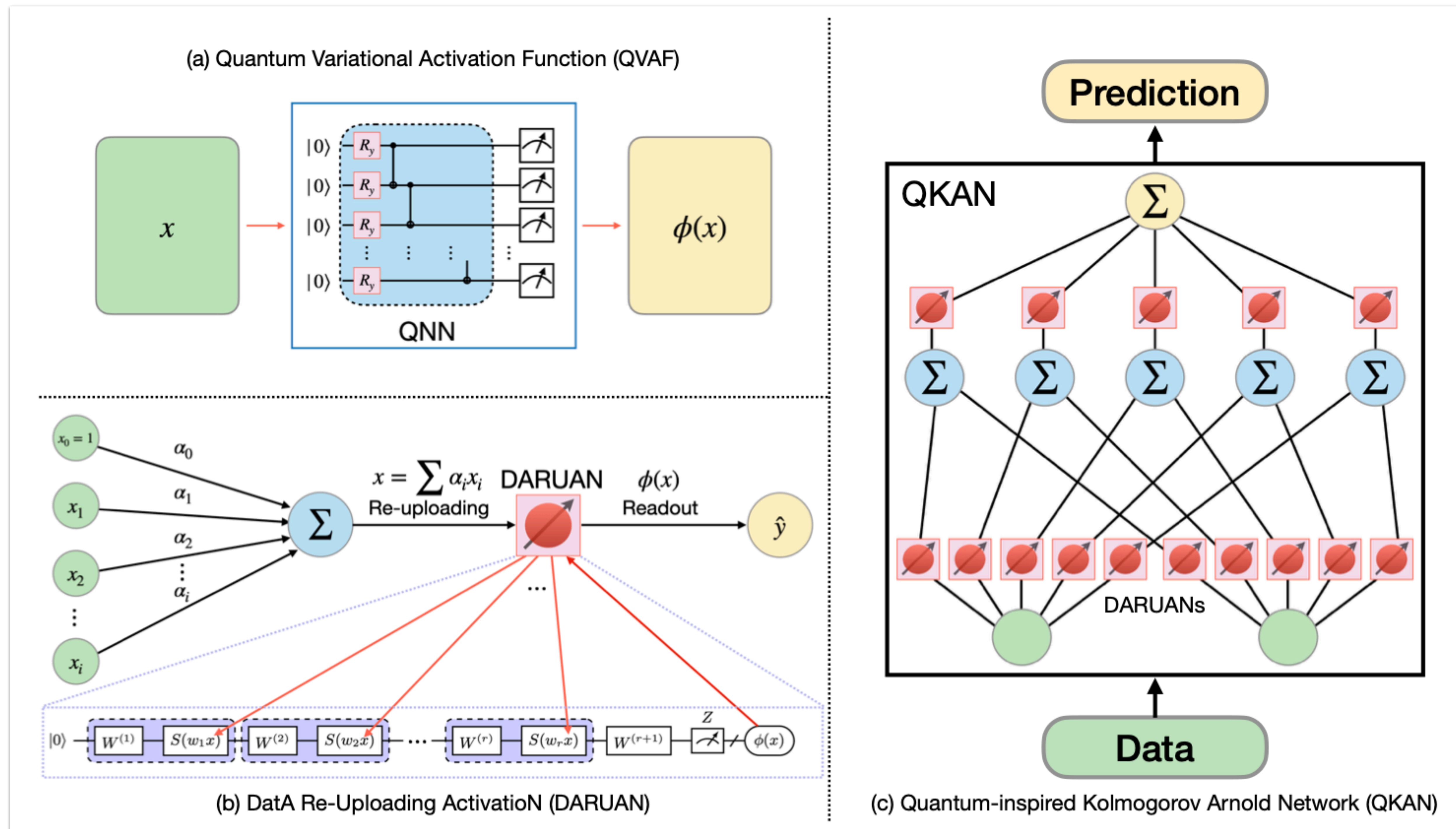
September 16, 2026 @ IEEE Quantum Week 2026

Scalable Validation and Optimized Simulation for Quantum Machine Learning — Tutorial



QKAN

Quantum-inspired Kolmogorov-Arnold Networks



KAN

What is Kolmogorov-Arnold Network?

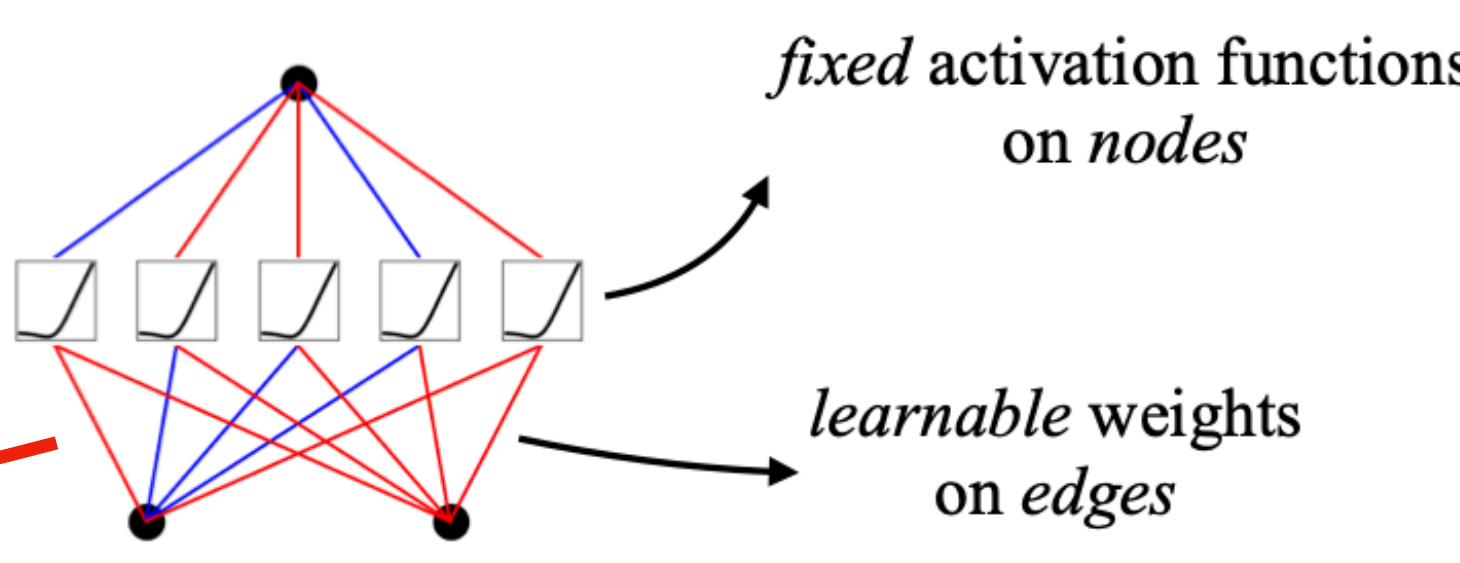
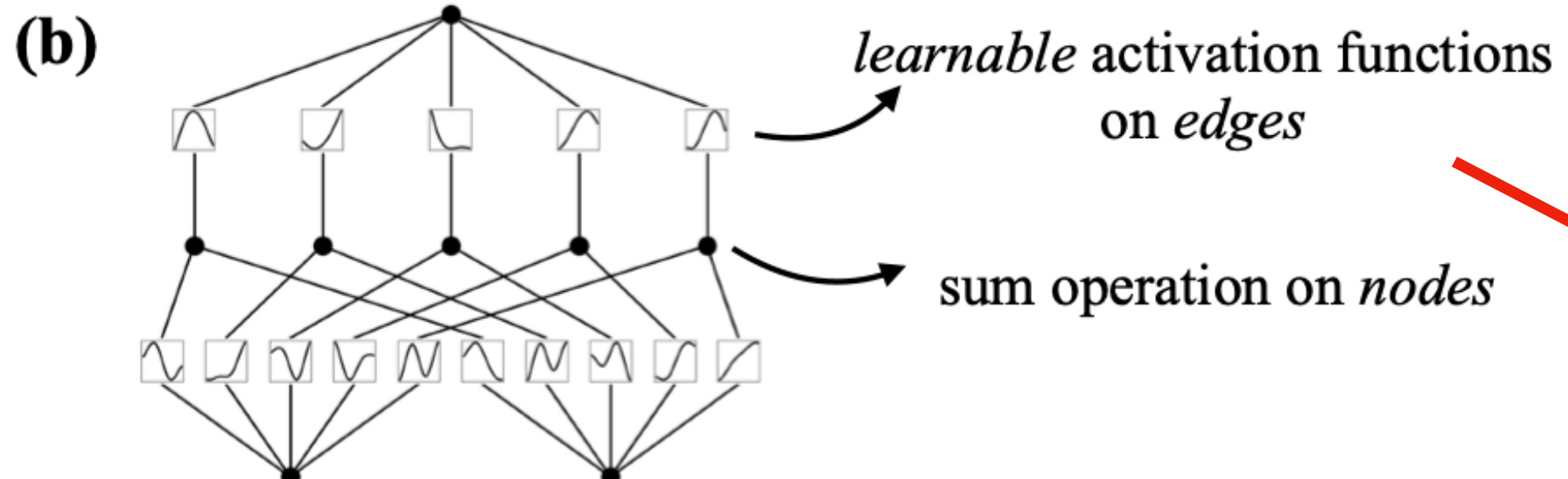
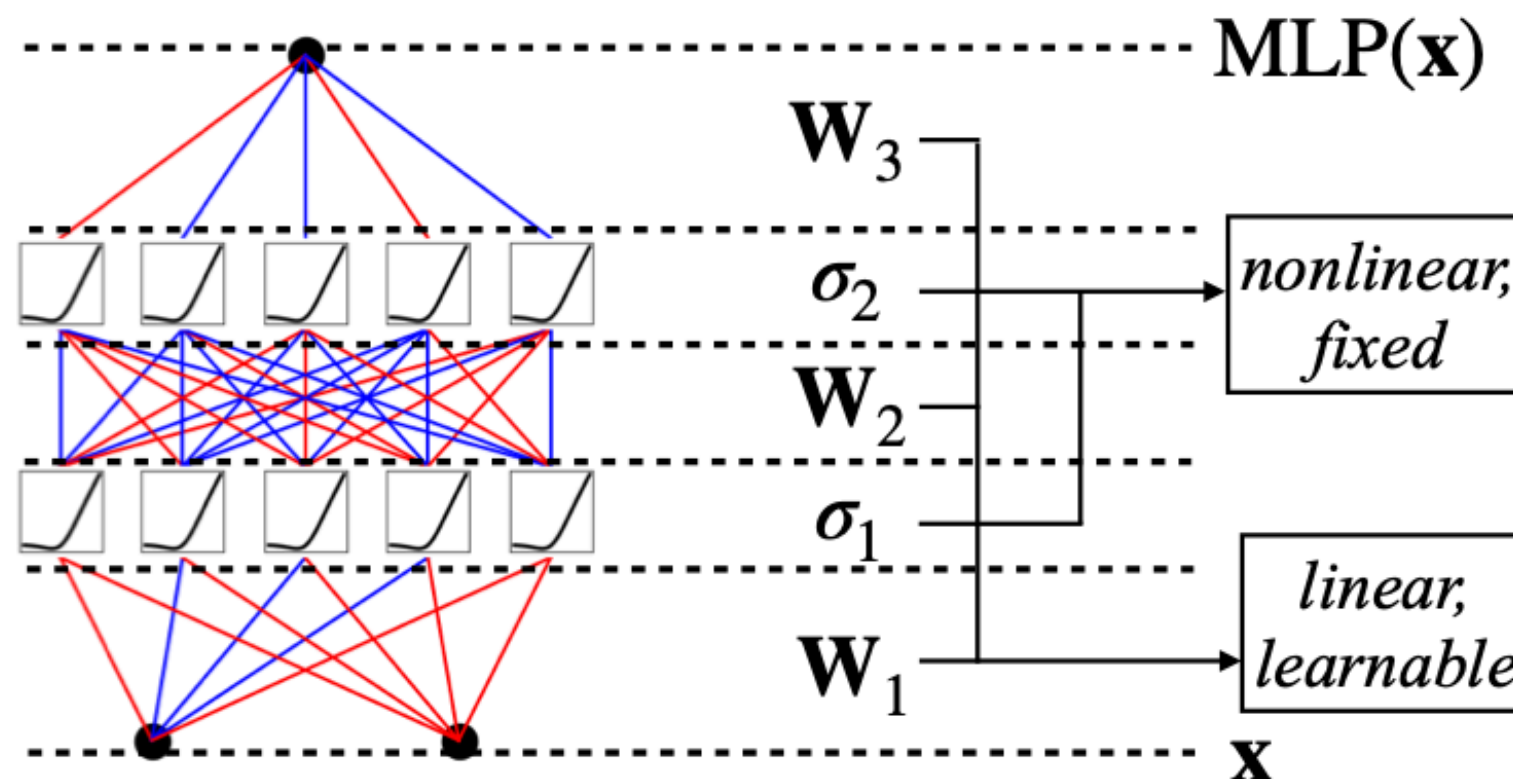
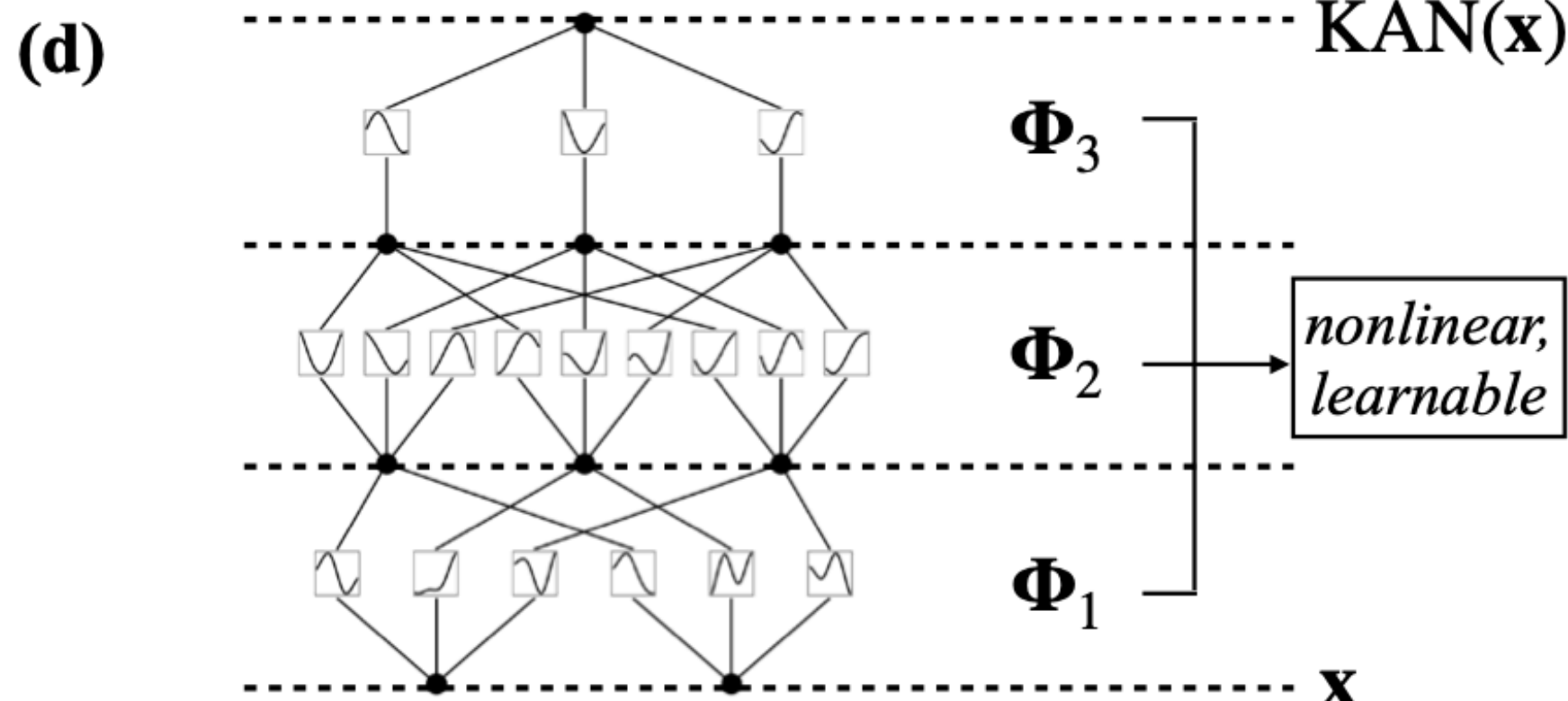
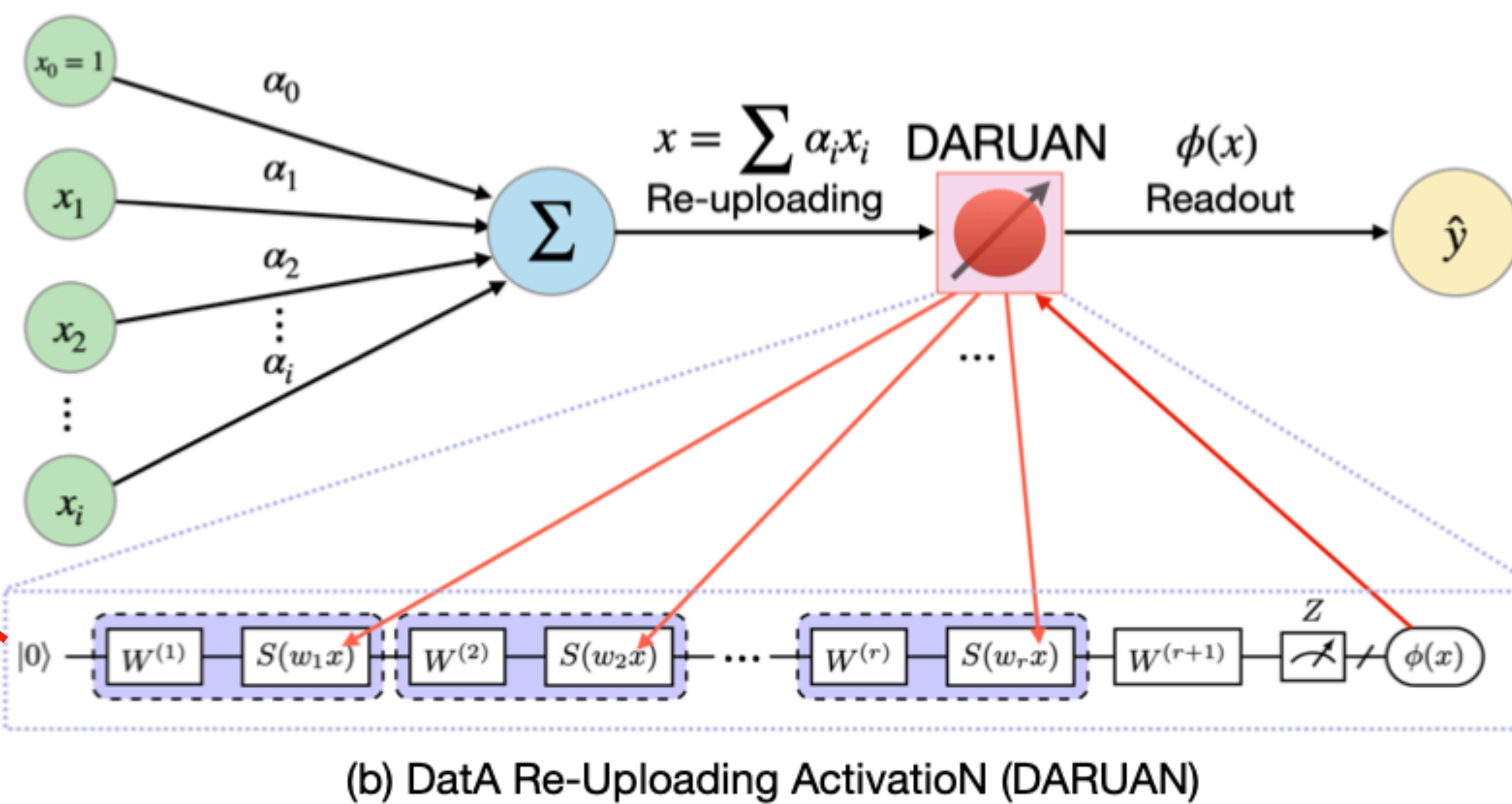
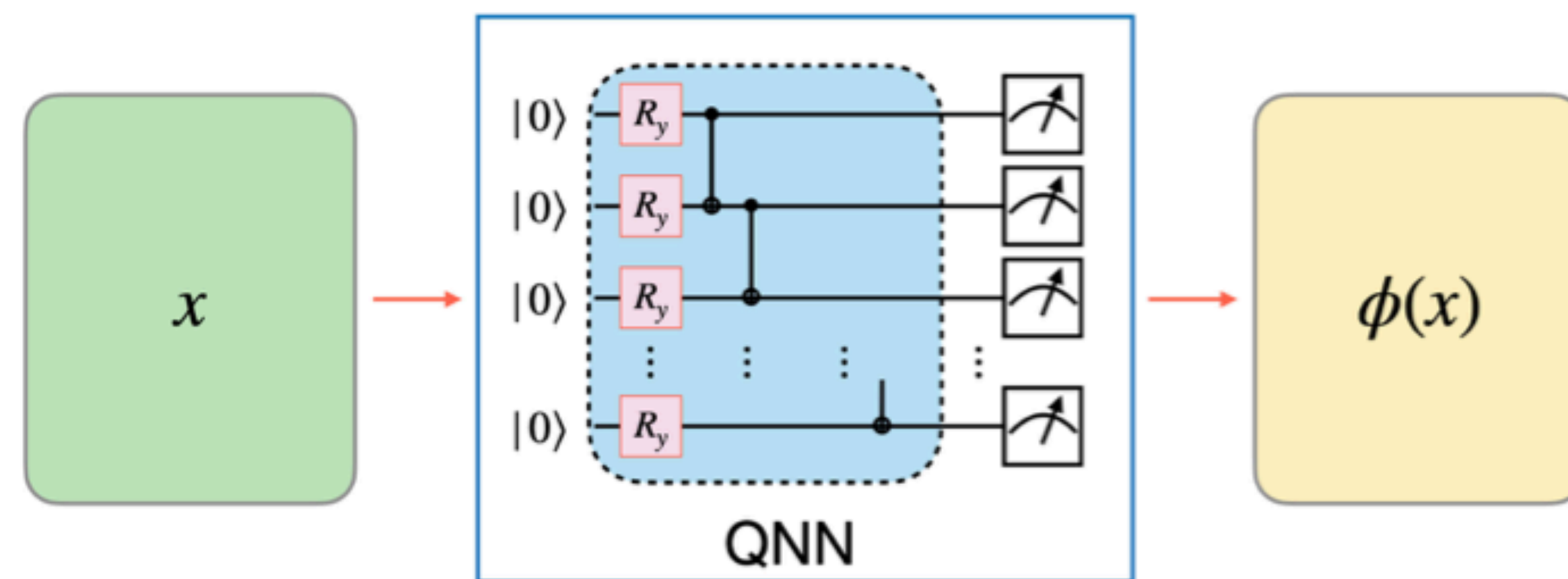
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a)  fixed activation functions on nodes learnable weights on edges	(b)  learnable activation functions on edges sum operation on nodes
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c)  MLP(x) $\mathbf{W}_3$ σ_2 $\mathbf{W}_2$ σ_1 $\mathbf{W}_1$ $\mathbf{x}$ nonlinear, fixed linear, learnable	(d)  KAN(x) Φ_3 Φ_2 Φ_1 $\mathbf{x}$ nonlinear, learnable

Figure 0.1: Multi-Layer Perceptrons (MLPs) vs. Kolmogorov-Arnold Networks (KANs)

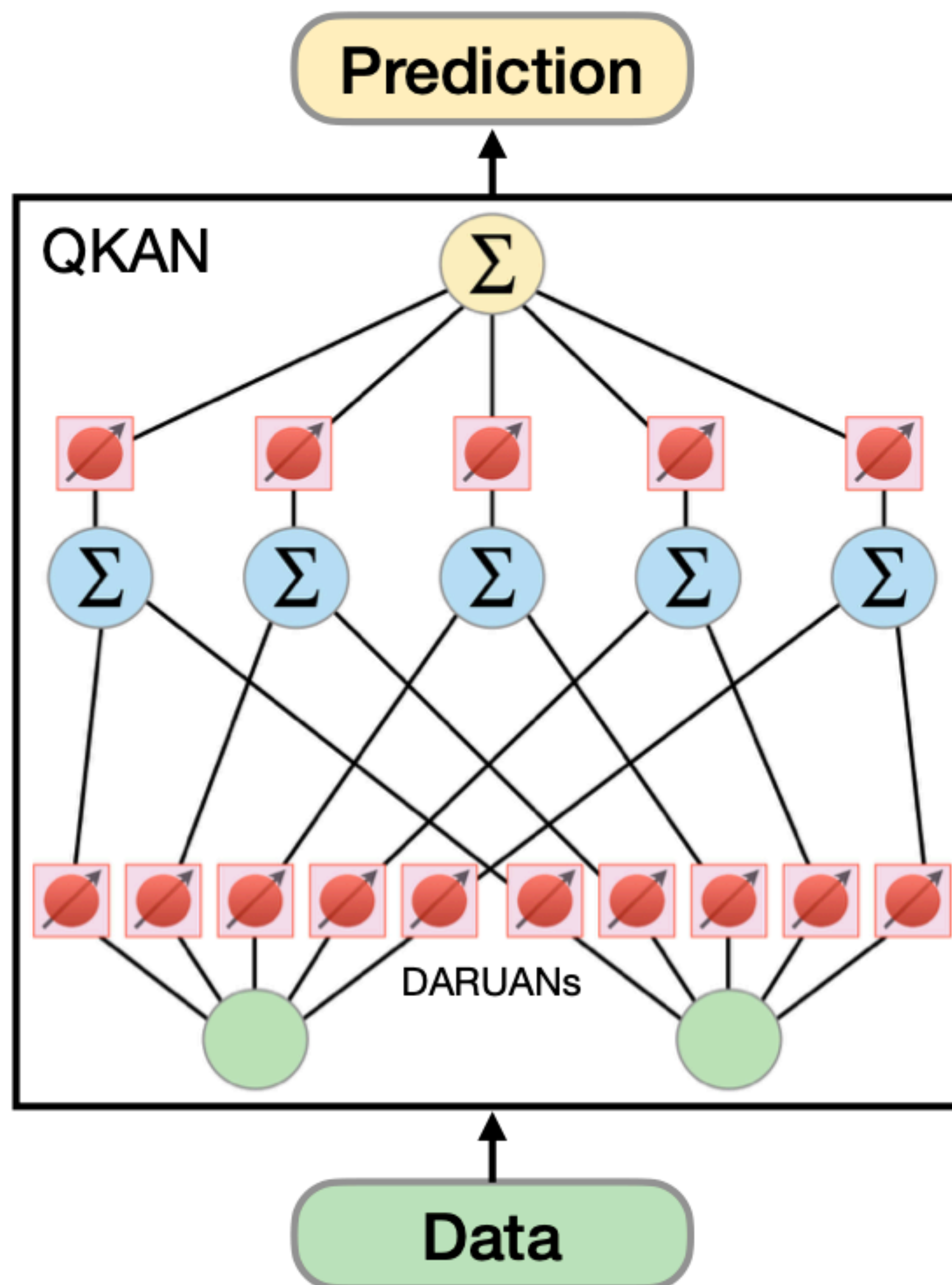
QKAN

Quantum-inspired Kolmogorov-Arnold Networks

(a) Quantum Variational Activation Function (QVAF)



Validation
w/ single qubit



(c) Quantum-inspired Kolmogorov Arnold Network (QKAN)

QKAN

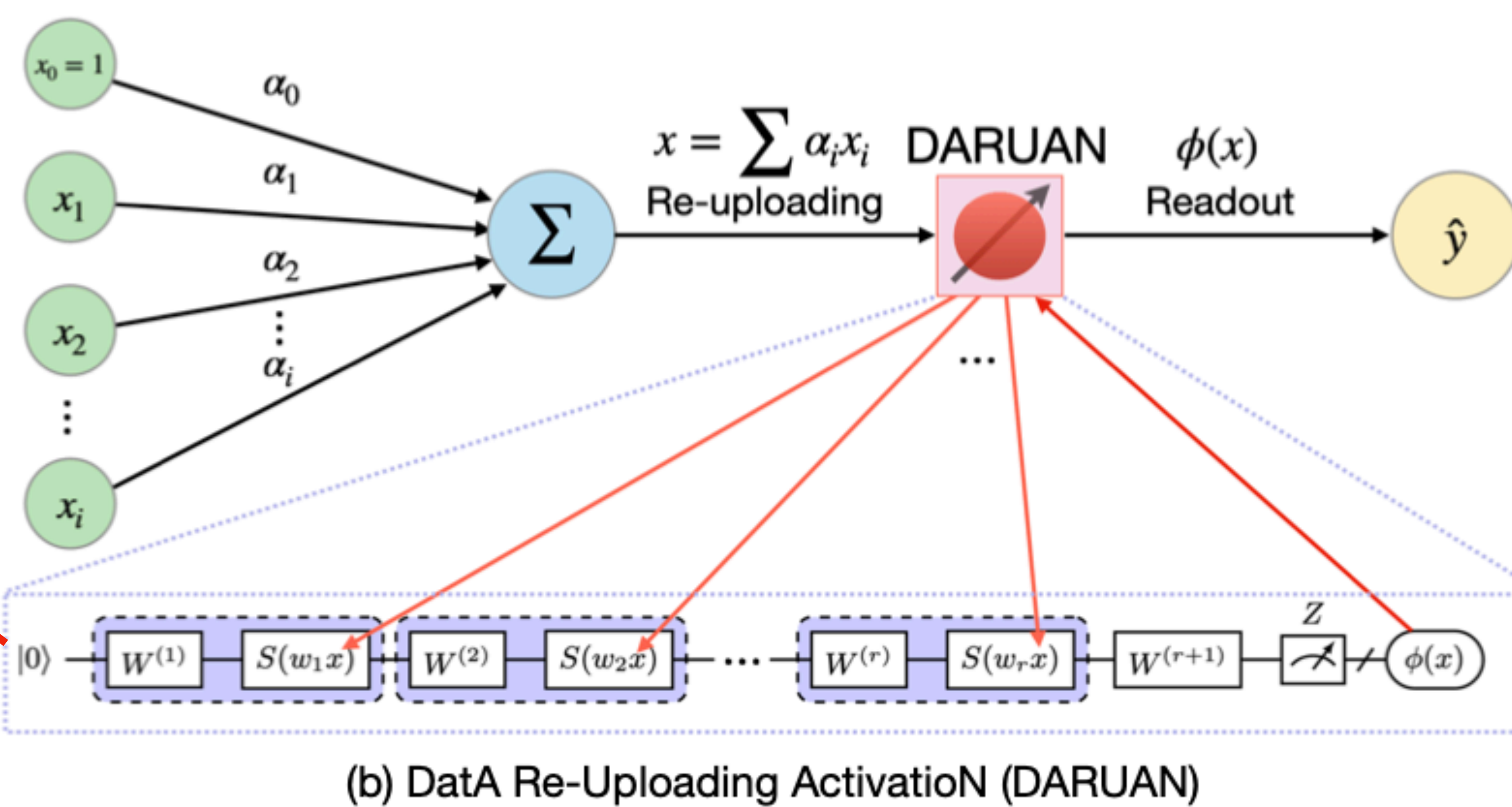
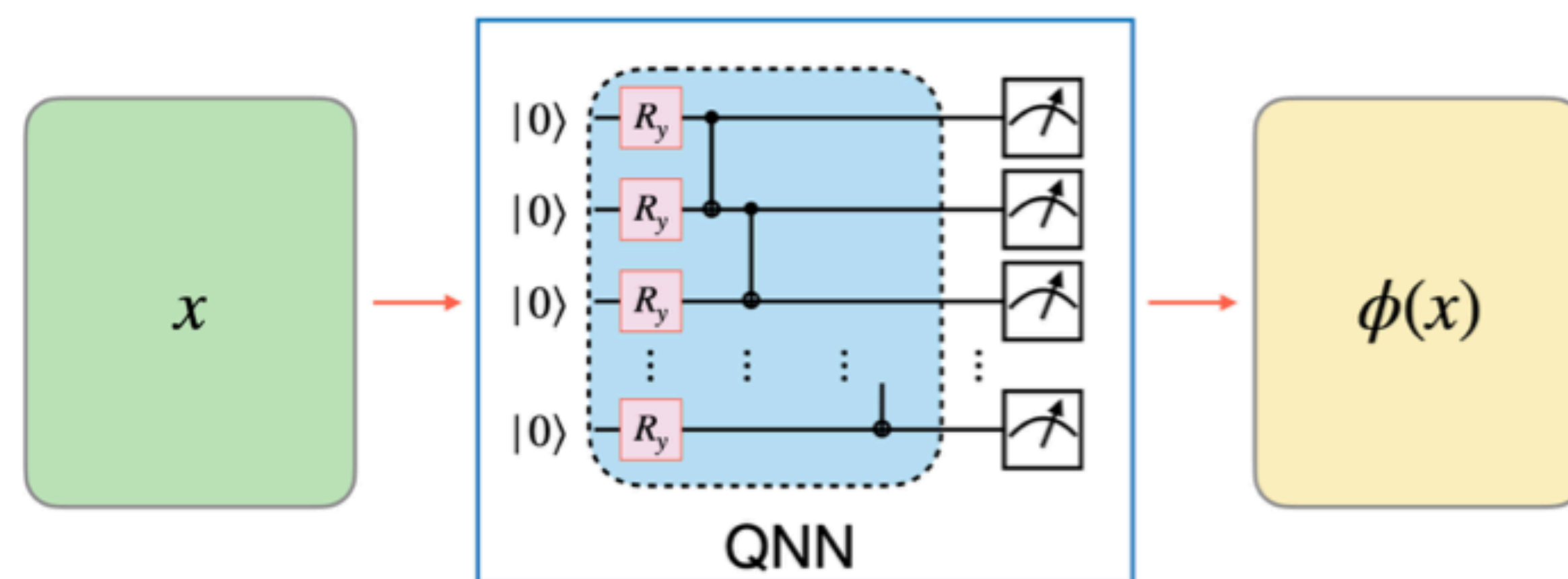
Logarithmically reducing parameter size

- Single qubit circuit used to show limited representation. (Like classical)
 - 1-qubit data re-uploading circuit has Fourier frequency spectrum size $|\Omega_A| = 2r$.
- In ours work, it shows that with data pre-processing weights in data encoding block, even 1-qubit circuit has **exponentially large** spectrum $1 \leq |\Omega_B| \leq (3^r - 1)$.
- We therefore obtained the parameter efficiency of QKAN compared to Fourier-KAN from $M = \Theta(\varepsilon^{-1/(k+1-m)})$ to $r = \left\lceil \frac{\log_2(C_f/\varepsilon)}{k+1-m} \right\rceil = \Theta(\log \frac{1}{\varepsilon})$, an **logarithmical reduction**.

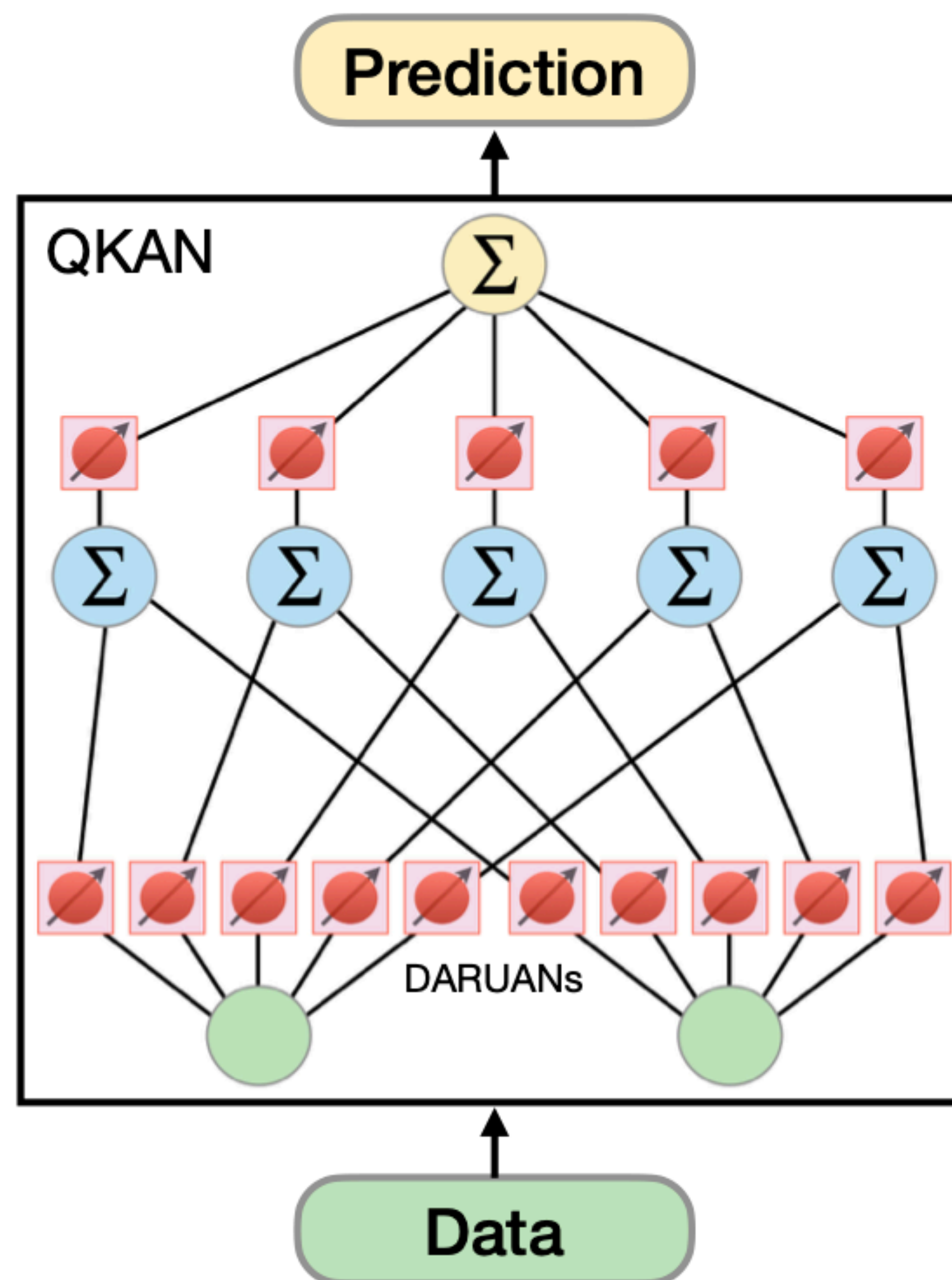
QKAN

Quantum-inspired Kolmogorov-Arnold Networks

(a) Quantum Variational Activation Function (QVAF)



(b) Data Re-Uploading Activation (DARUAN)

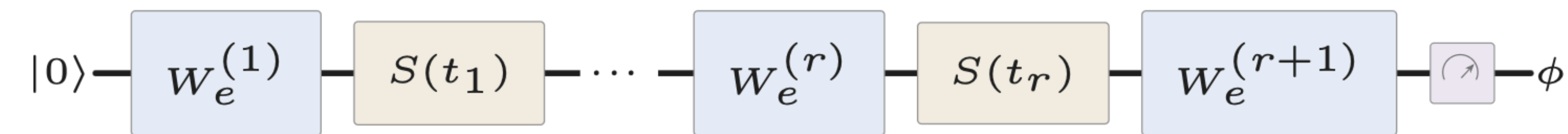


(c) Quantum-inspired Kolmogorov Arnold Network (QKAN)

FlashQKAN

IO-Aware Parallel Single-Qubit Simulation

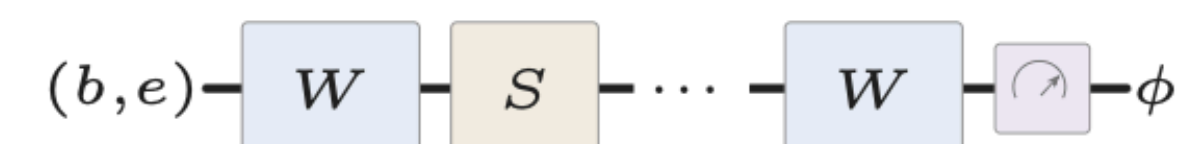
(a) One-edge quantum activation



$$t_{b,e,\ell} = w_{e,\ell} X_{bi} + \beta_{e,\ell}$$

packet (e, T) : reuse p_e , vary $X_{T,i}$

(b) Reference exact replay



materialize gates

state updates

save tape

HBM
gates
states
tape

repeated HBM traffic

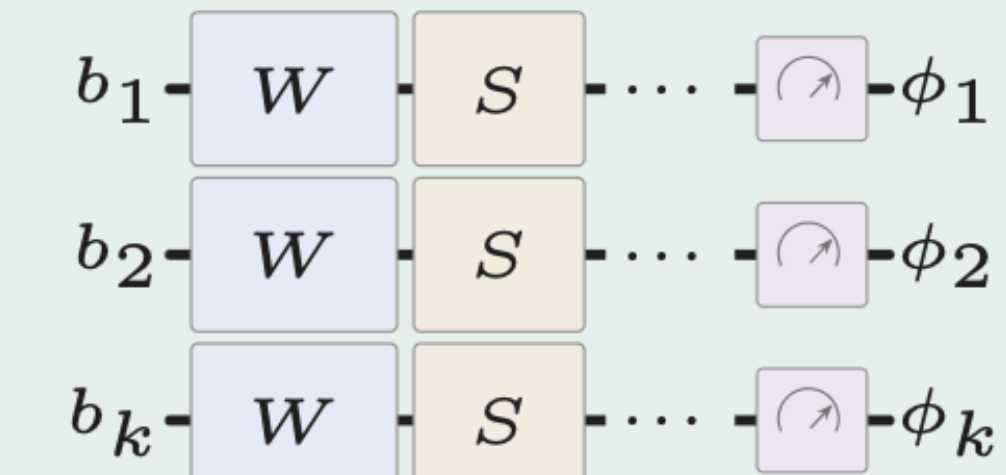
(c) FlashQKAN packet simulation

load $X_{T,i}$

load p_e

write $H_{T,j,i}$

on-chip packet



$$\psi_T \leftarrow S(t)W\psi_T$$

one launch; one HBM write

FlashQKAN

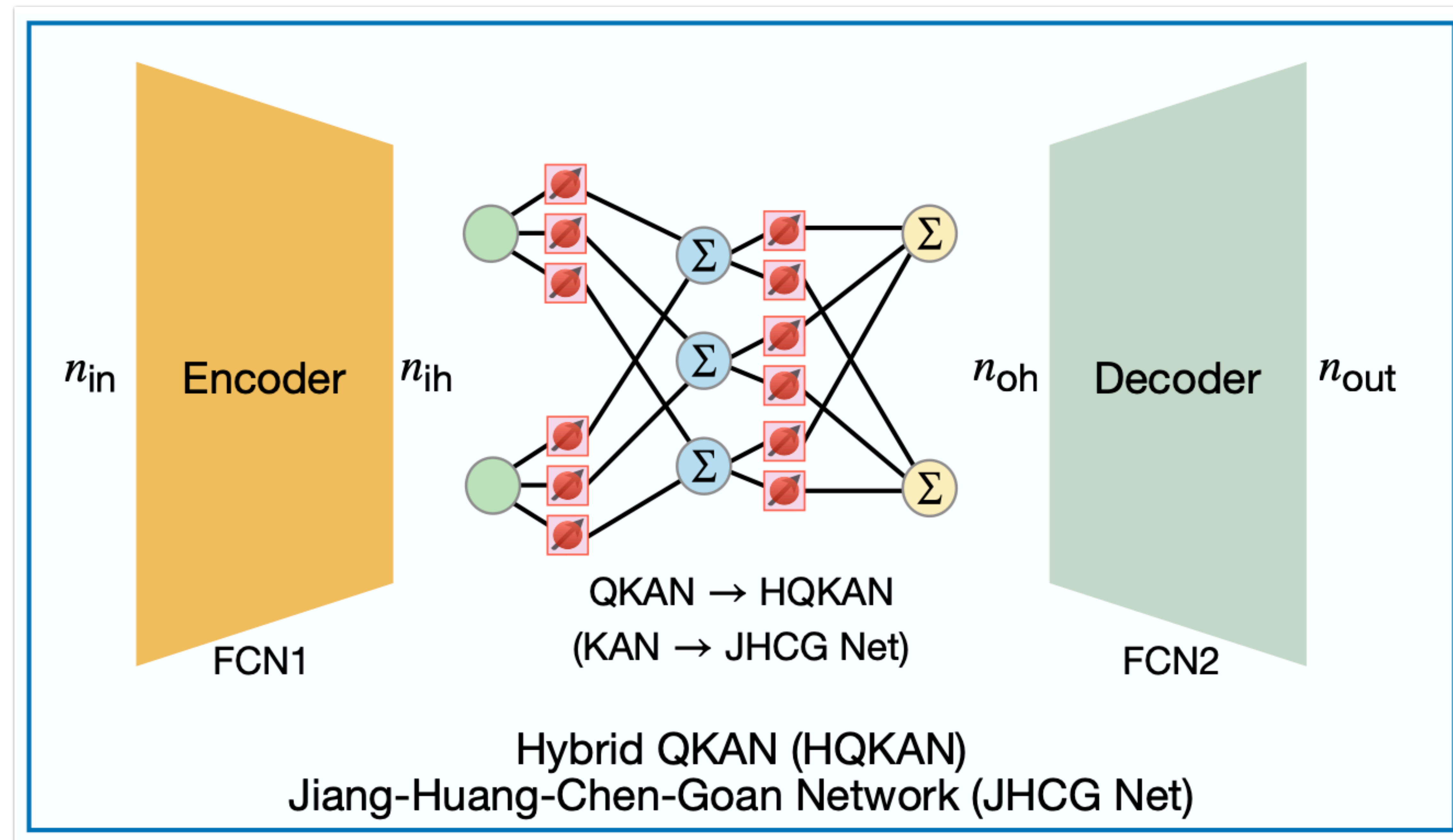
IO-Aware Parallel Single-Qubit Simulation

- Accelerated through tile-based program, cuTe DSL, with 100x speed up.

		RTX 5090				H200 SXM			
Ansatz	Solver	F+B ms	Speedup	Peak MB	Max err.	F+B ms	Speedup	Peak MB	Max err.
pz	exact	17.803	1.00×	229.4	0	5.999	1.00×	229.4	0
pz	flash	0.206	86.6×	200.0	1.62e-6	0.353	17.0×	200.0	2.13e-6
pz	cute	0.290	61.4×	200.0	3.16e-6	0.223	26.9×	200.0	3.16e-6
rpz	exact	9.433	1.00×	291.9	0	18.646	1.00×	291.9	0
rpz	flash	0.282	33.5×	136.0	1.27e-6	0.347	53.8×	136.0	1.40e-6
rpz	cute	0.167	56.4×	136.0	2.87e-6	0.203	91.9×	136.0	3.63e-6
real	exact	8.242	1.00×	76.7	0	2.877	1.00×	76.7	0
real	flash	5.186	1.59×	168.0	2.61e-6	5.865	0.49×	168.0	2.61e-6
real	cute	0.262	31.5×	168.0	2.91e-6	0.205	14.0×	168.0	2.90e-6

Hybrid QKAN

Further reducing parameter in novel architecture

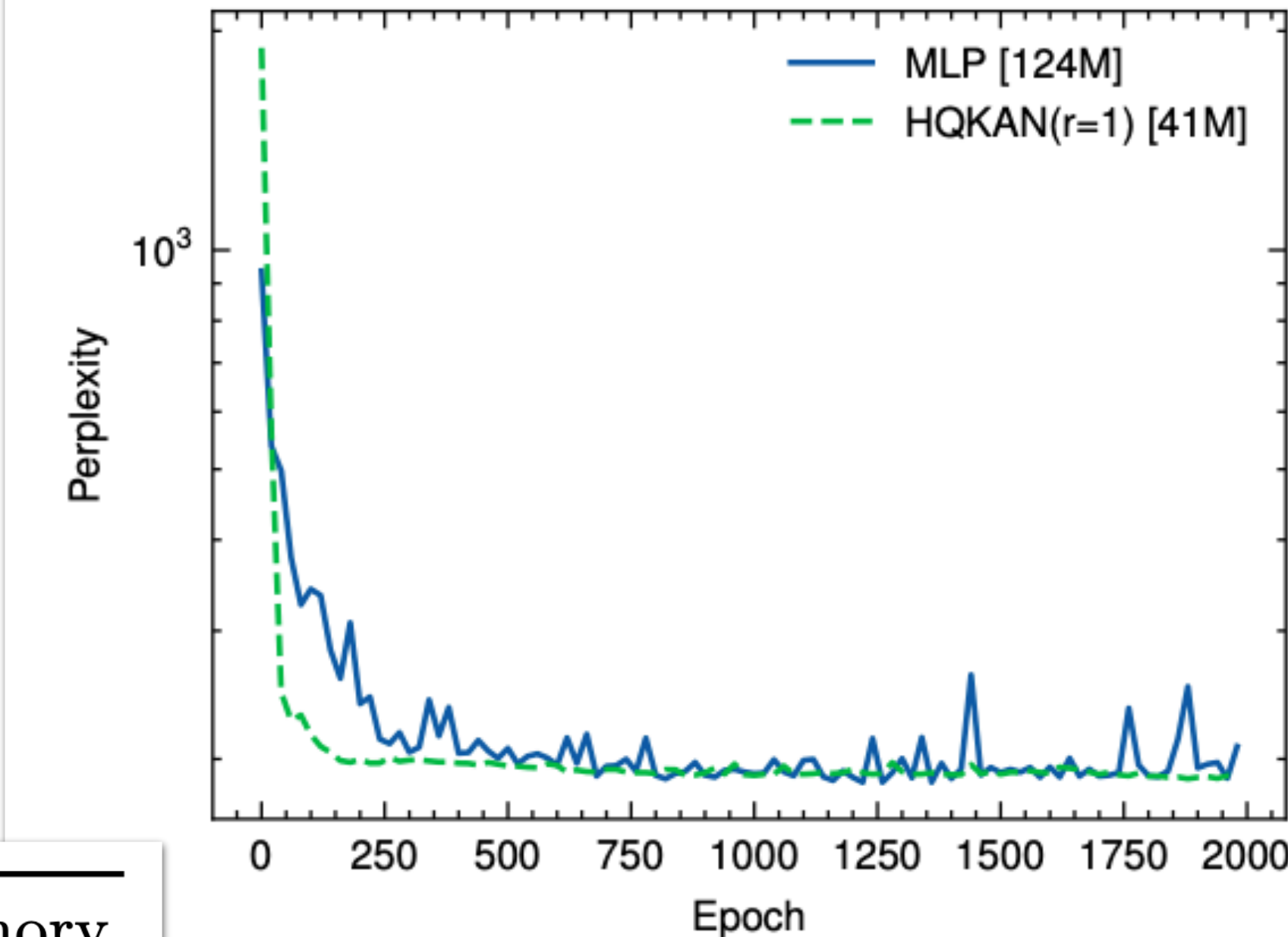


Hybrid QKAN

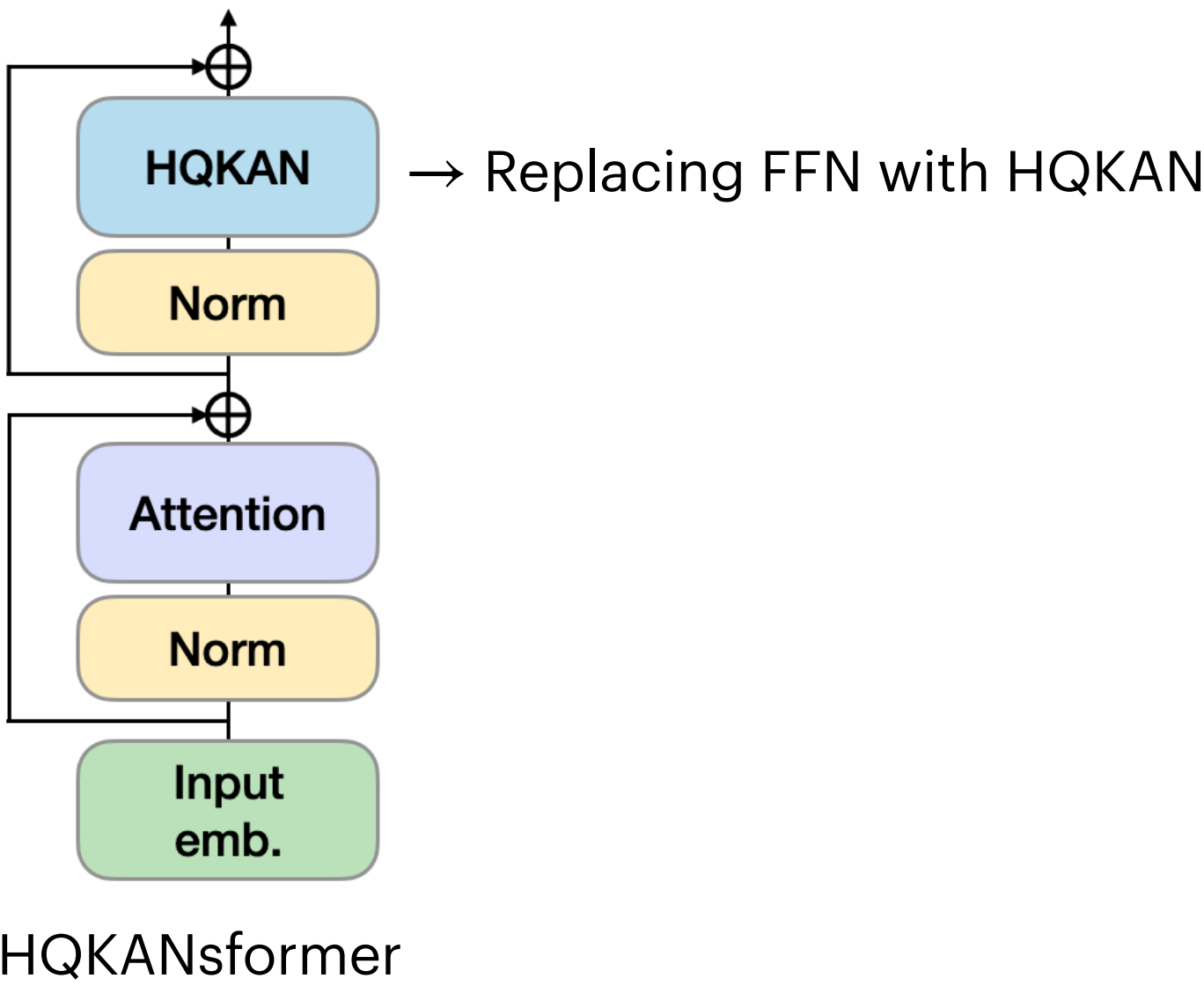
Outperform classical model in both accuracy and param size

Table 3: Performance of different models on MNIST, CIFAR-10, and CIFAR-100 datasets. The top-1, top-5 test accuracy and the parameter size of each model. The second column indicates the parameter size of CNN shared by all models.

Model	(CNN)	CNN+MLP			CNN+KAN(g=10)			CNN+QKAN(r=3)			CNN+HQKAN		
Training dataset	CNN # Params	Top-1 Accuracy (%)	Top-5 Accuracy (%)	MLP # Params	Top-1 Accuracy (%)	Top-5 Accuracy (%)	KAN # Params	Top-1 Accuracy (%)	Top-5 Accuracy (%)	QKAN # Params	Top-1 Accuracy (%)	Top-5 Accuracy (%)	HQKAN # Params
MNIST	1,084	97.9	100.0	850	97.7	100.0	1,500	98.0	100.0	800	95.9	99.7	222
CIFAR-10	56,320	71.4	97.8	41,802	68.4	97.4	39,900	68.8	97.0	21,280	71.6	97.9	14,370
CIFAR-100	56,320	39.8	69.4	86,948	40.6	70.4	384,000	41.2	70.0	204,800	39.9	70.6	32,636



Method	# Params	Batch size	Time/iter [GPU · ms]	Device memory
MLP-based transformer w/ flash atten.	124 M	1	40 [RTX 4090 · ms]	4.7 GB
		10	252 [RTX 4090 · ms]	17.3 GB
		48	3,884 [V100S · ms]	81.2 GB
		320	6,400 [H100 · ms]	656 GB
		800	14,784 [H200 · ms]	1,340 GB
HQKANsformer w/ flash atten. 	67 M	1	39.5 [RTX 4090 · ms]	3.8 GB
		10	240 [RTX 4090 · ms]	15.6 GB
		48	3,540 [V100S · ms]	73.6 GB
		320	5,760 [H100 · ms]	592 GB
		800	13,232 [H200 · ms]	1,224 GB



Hybrid QKAN on LLM

Chat results from pre-training and knowledge distillation

1.8B MLP

Imagine that you walk into a cafe to have a cup of coffee.
Describe the experience in less than 100 words.

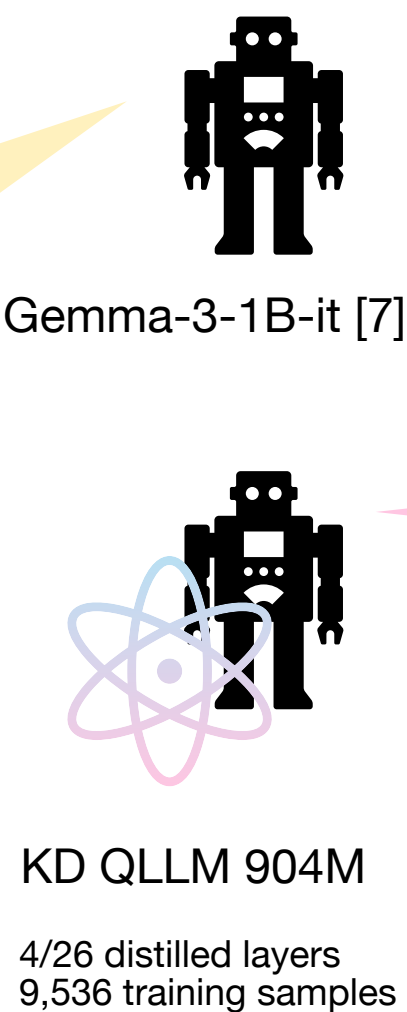
We walk in, a gentle hum of the coffee machine and the soft chatter of patrons. A steaming cup of coffee is placed on each small table, each inviting us to sit and enjoy. The aroma of freshly ground coffee wafts through the air, enticing our senses. The plush carpet beneath our feet is inviting, and the soft hum of the coffee machine is a background noise. Our eyes wander to the steaming cups, each one in a different shade of brown, inviting us to take one and savor the flavors. The sounds of the café are a gentle reminder that it's a place for connection and community.

800M QKAN

Imagine that you walk into a cafe to have a cup of coffee.
Describe the experience in less than 100 words.

I arrive at the coffee shop, feeling a bit overwhelmed by the bustling atmosphere. The air is thick with the smell of coffee, roasting from a single pot spread throughout the cafe. A lone barista greets me, a warm smile spreading across their face. They expertly mix and brew the perfect cup of coffee, their hands moving deftly with a practiced air. Every sip is a labor of love, pouring out the best that this small space has to offer. I take a sip, feeling the rich aroma fill my senses, and savor the warmth of the coffee.

The rain hammered against the windows, a grey, comforting scene. The cafe smelled of dark chocolate and woodsmoke. Stepping inside, the warmth of the mugs felt welcoming. A gentle hum of conversation drifted from the corner, and the light filtering through the window painted the walls with an inviting glow. Just a simple moment, a quiet refuge from the storm, sipping coffee and feeling utterly peaceful.

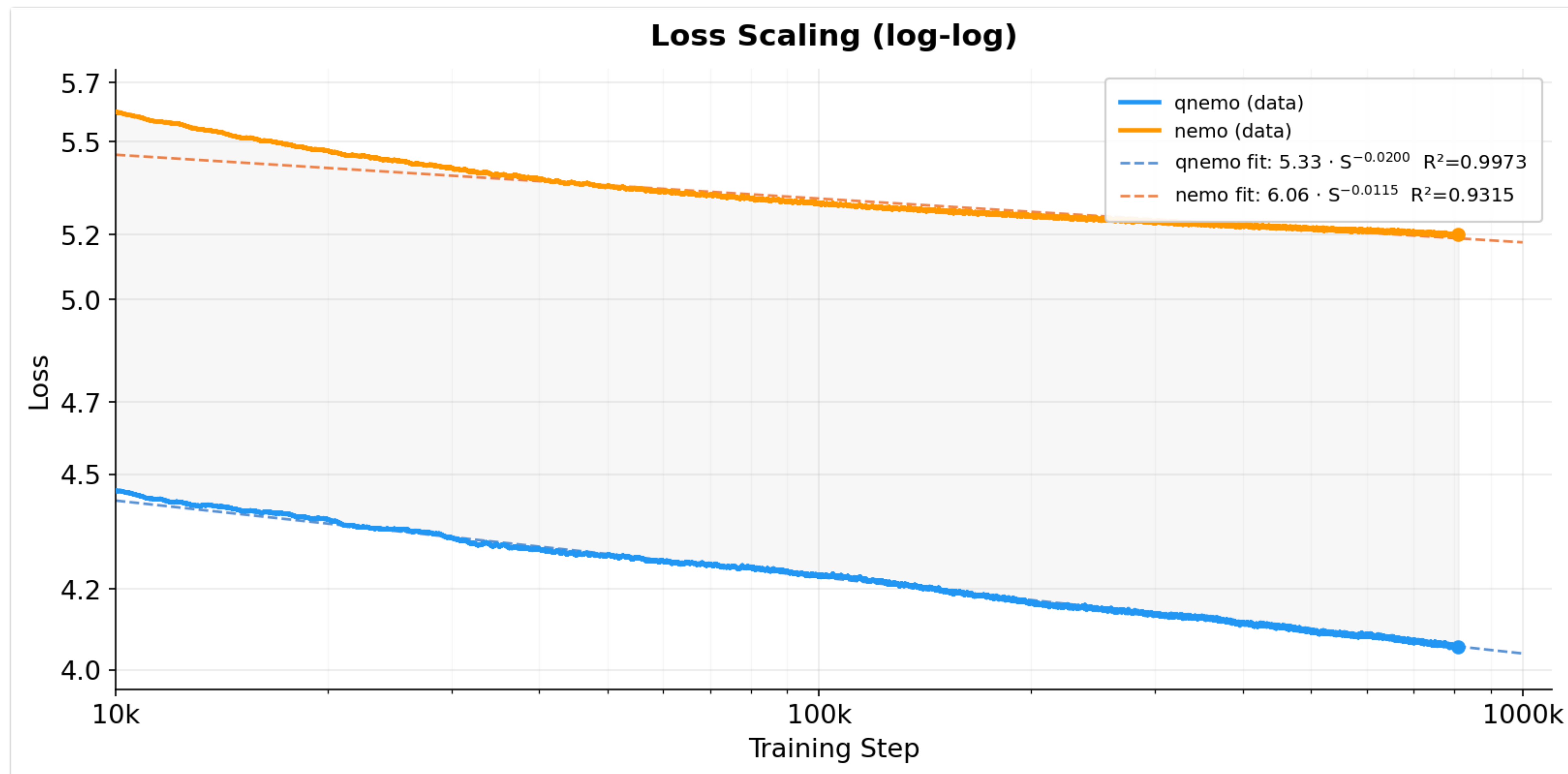


Sunlight filtering through windows illuminate the cozy cafe. The scent of coffee fills the air, combined with sweet chocolate and cinnamon. A quiet hum of conversation fills the space, contrasted by the gentle clink of coffee mugs. I sit down on a cushioned seat, tucking my lunch into my bag. As I sip the warm coffee, the world momentarily stops, and the moment feels both comforting and peaceful. It's just a good start to a day.

Scaling Law

Nemotron-flash-3b-QKAN-1.6B vs Nemotron-flash-3B on 300B tokens

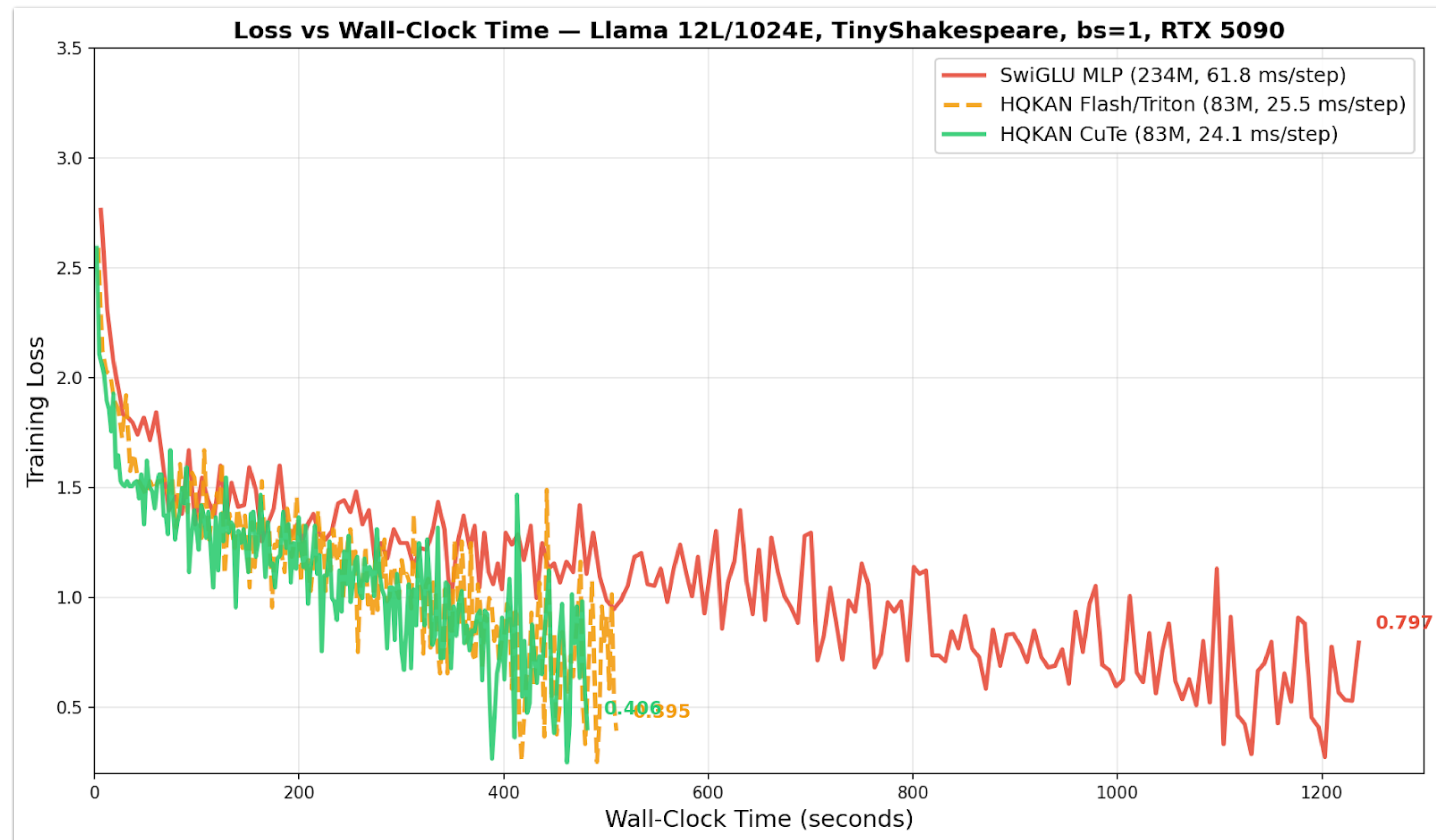
- Two times larger than classical in exponent part!



Faster than classical

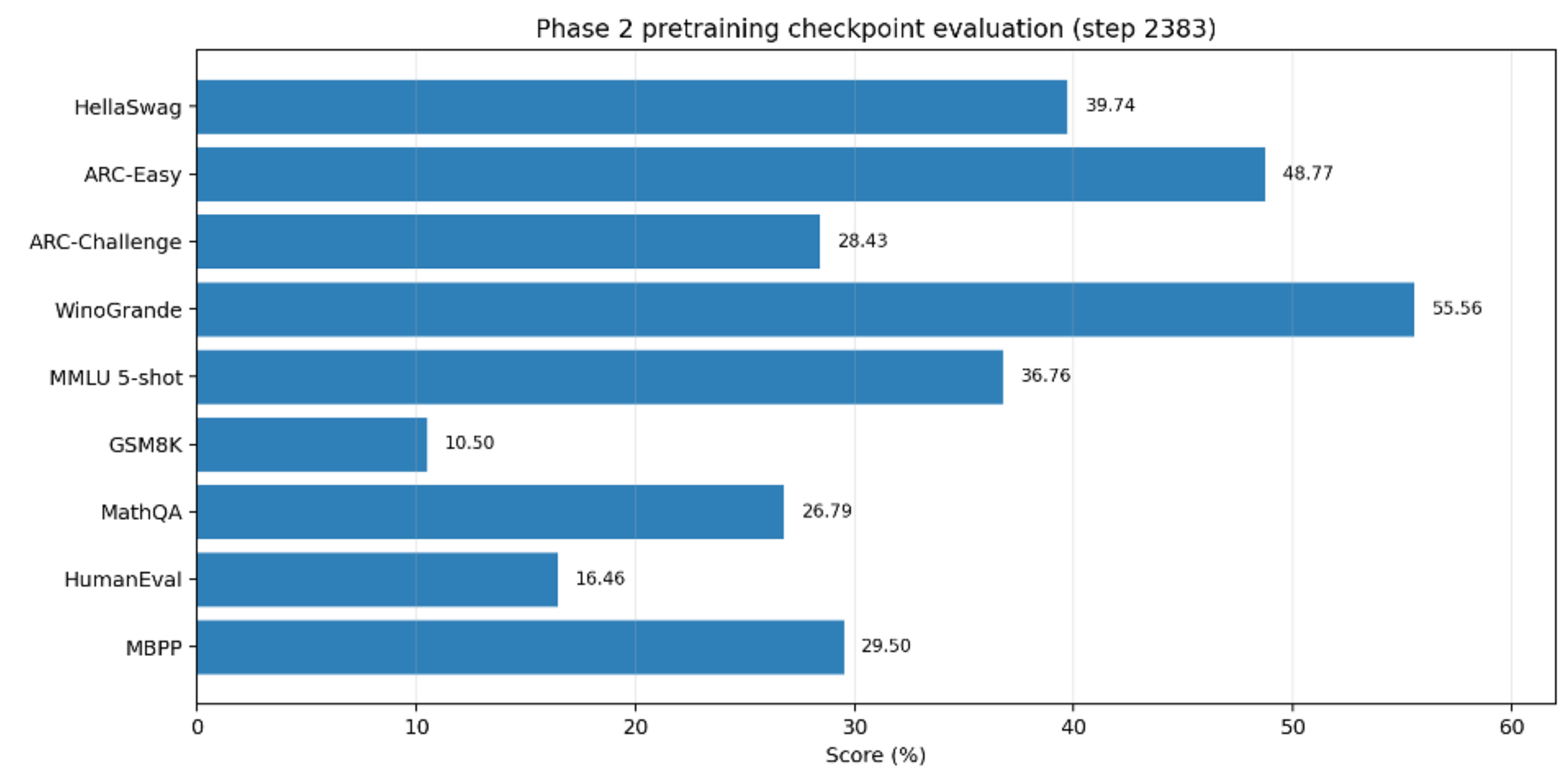
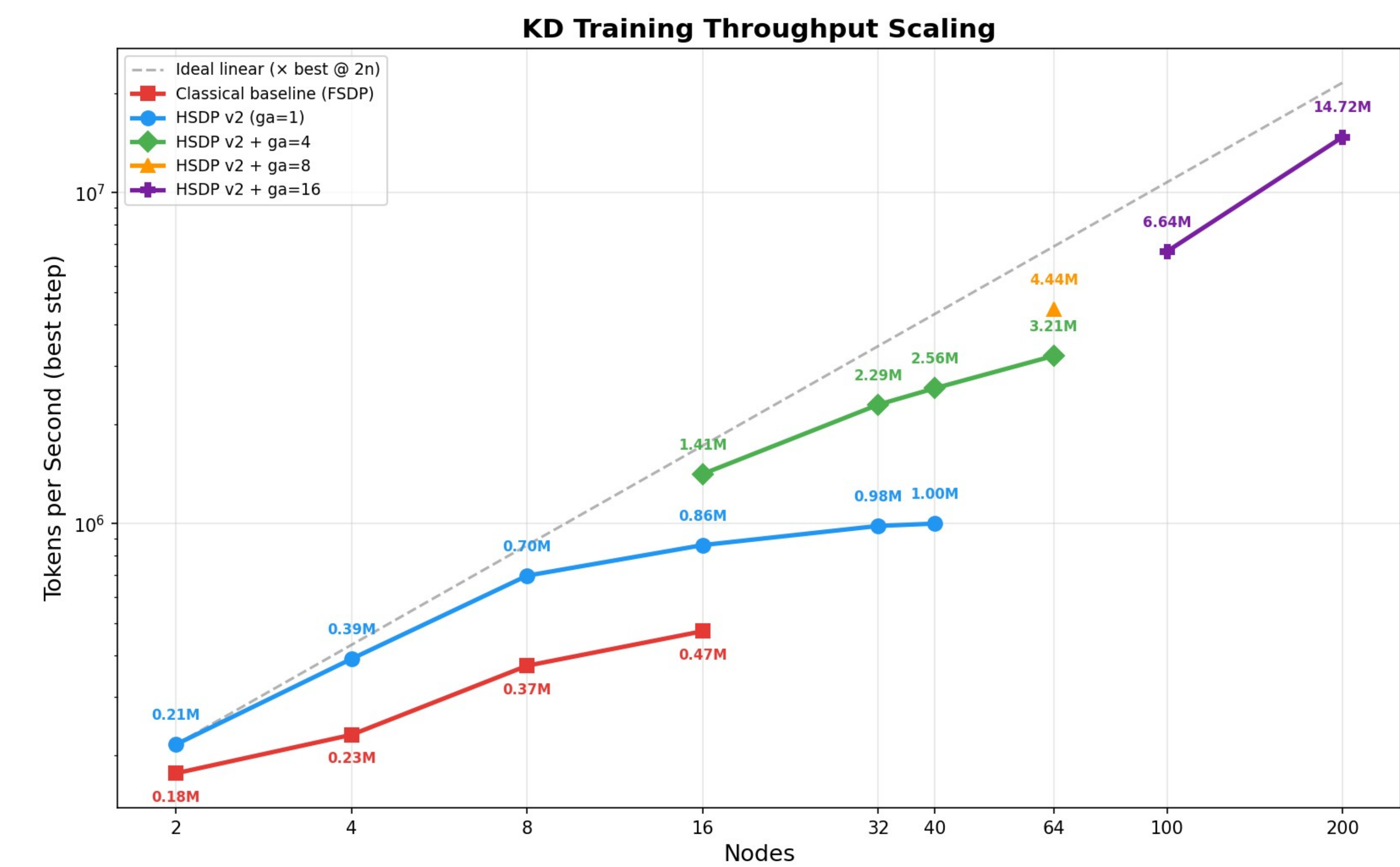
Wall time comparison in Llama model under 2000 steps

- Around 3x faster than classical model.



Nemotron-flash-3b-QKAN-1.6B KD from Nemotron-nano-9b-v2

Scaling up in NCHC Infrastructure Nano4



QKAN applications in Quantum Week

Gated QKAN-FWP: Scalable Quantum-inspired Sequence Learning

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Chen-Yu Liu^{6,}, En-Jui Kuo^{7,}, Yun-Yuan Wang^{4,}, Prayag Tiwari^{8,}, Andrea
Ceschini^{9,}, Chi-Sheng Chen^{10,}, Yu-Chao Hsu^{2,11,}, Chun-Hua Lin^{1,2,}, Tai-Yue
Li^{2,}, Antonello Rosato^{9,}, Massimo Panella^{9,}, Simon See^{12,}, Saif Al-Kuwari^{13,},
Kuan-Cheng Chen^{†,13,}, Nan-Yow Chen^{†,2,}, Hsi-Sheng Goan^{†,1,5,6,14,}

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June 16, 2026

QML track, Thu 10 am, 803A

Generative Quantum-inspired Kolmogorov-Arnold Eigensolver

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Chun-Hua Lin^{2,5,}, Kuo-Chung Peng^{2,5,}, Jiun-Cheng Jiang^{5,6,7,}, Yun-Yuan Wang^{6,},
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QGDD track, Thu 3 pm, 705

Hybrid Quantum-inspired Kolmogorov–Arnold Networks for Privacy-Aware Federated Biosignal Learning

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PRQAI wks paper



